

Kill Sheet — Surface

The IWCF Surface BOP vertical-well kill sheet, calculated live

⚠ FOR UNDERSTANDING PURPOSES ONLY

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Flutter / Dart

IWCF

Surface BOP

8 live calculations

2-page PDF

1. What It Is

A **kill sheet** is the main document used to control a well during a kick. It works out the exact pressures and pump strokes the driller must follow to circulate the influx out and replace the mud with heavier **kill weight mud**.

This screen turns the IWCF (International Well Control Forum) paper form, “Surface BOP Vertical Well Kill Sheet”, in API field units, into a live calculator. Every labelled box on the paper form, (A), (B), (C) and so on, has its own result on screen. It is the most complex screen in the well-control section: all eight calculations run as the driller types, and it produces a closely matching IWCF-style A4 PDF.

Standard	IWCF Surface BOP, vertical well, API field units
Reached from	Operations › Drilling › Well Control › Kill Sheets › Surface
Saving	Five named slots in the app database
Output	Two-page A4 IWCF-formatted PDF through the Share menu
Extras	A well diagram and a step-down pressure chart, both drawn by the app

2. The Seven Sections

#	Section	Contents
1	Formation Strength Data	Formation strength test (A), mud weight at test (B), maximum allowable mud weight (C, automatic), initial MAASP (automatic)
2	Current Well Data	Mud density, casing shoe data (size, MD, TVD), hole data (size, MD, TVD) and a well diagram
3	Pump Data	Pump displacement, strokes per minute and slow-pump pressure

#	Section	Contents
4	Volumetric Data & Pump Strokes	Drill string, open-hole annulus, casing annulus and surface volume, with totals (D) to (J)
5	Kick Data & Kill Mud Weight	SIDPP, SICP, pit gain and kill mud weight (automatic)
6	Circulating Pressures	Slow-pump pressure, initial circulating pressure, final circulating pressure and K100
7	Drill Pipe Pressure Step-Down	A schedule and line chart of how pressure should fall as kill mud fills the string

IWCF letters

(A) formation strength · (B) mud weight at test · (C) maximum allowable mud weight · (D) drill string total · (F) open-hole annulus total · (G) casing annulus total · (H) total annulus · (I) total well system · (J) surface volume.

3. How to Use It

1. Enter the **formation strength** and mud weight at test. The maximum allowable mud weight and the initial MAASP appear.
2. Enter the **current well data**. The well diagram updates with your depths.
3. Enter the **pump data**: displacement, speed and slow-pump pressure.
4. Fill in drill string, open-hole annulus and casing annulus lengths and capacities, then the active surface volume. Volumes, strokes and circulation times all calculate.
5. Enter **SIDPP, SICP and pit gain**. The kill mud weight appears.
6. The **circulating pressures** update once the kill mud weight is known.
7. The **step-down schedule** generates itself, every 100 strokes, with a line chart.
8. **Save, Open or Export PDF**.

4. The Eight Calculations

They always run in dependency order, so every formula has its inputs ready. They re-run whenever any value changes or a saved slot is opened.

Calculation	How it is worked out
1. Volumes	For each row, volume (bbl) = length (ft) × capacity (bbl/ft). Rows add up to the totals (D) to (J)
2. Strokes and times	Strokes = volume ÷ pump displacement; time (min) = strokes ÷ strokes per minute. Both always round up

Calculation	How it is worked out
3. Max allowable mud weight (C) and MAASP	$(C) = \text{mud weight at test} + \text{test pressure} \div (\text{shoe TVD} \times 0.052)$, rounded down to 0.1 ppg. $\text{MAASP} = ((C) - \text{current mud weight}) \times \text{shoe TVD} \times 0.052$, rounded down to a whole number
4. Kill mud weight (KMW)	$\text{Current mud weight} + \text{SIDPP} \div (\text{hole TVD} \times 0.052)$, rounded up to the next 0.1 ppg
5. Drill pipe pressure at slow rate (DPL)	The recorded slow-pump pressure
6. Initial circulating pressure (ICP)	DPL + SIDPP
7. Final circulating pressure (FCP) and K100	$\text{FCP} = (\text{KMW} \div \text{current mud weight}) \times \text{DPL}$. $\text{K100} = (\text{ICP} - \text{FCP}) \div \text{drill-string strokes} \times 100$
8. Step-down schedule	Starting at ICP, the pressure drops by K100 every 100 strokes, never going below FCP

Rounding is a safety decision

Strokes, times and the kill mud weight are always rounded **up**; the maximum allowable mud weight and MAASP are always rounded **down**. Every rounding errs on the side of safety. The constant 0.052 converts pounds-per-gallon $\times$ feet into psi.

5. Saving

Surface kill sheets use the app's shared database with **five named slots**. The slot label defaults to the sheet date and can be edited. Saving packs every field on the screen into the slot; opening a slot refills every field and re-runs all eight calculations. Saved sheets survive reinstalls and device backup and restore.

6. The PDF

The PDF closely reproduces the paper IWCF form across two A4 pages, with a dense layout to match the field document.

PDF section	Contents
Page header	IWCF title with date and name fields
Formation strength block	Rows (A) to (C) and MAASP
Pump data block	Displacement, speed and pressure cells
Current well data and diagram	Data labels plus the well image with rotated TVD labels
Volume and stroke table	Lettered rows (D) to (J) with volume, strokes and time
Kick data and KMW	SIDPP, SICP, pit gain and kill mud weight

PDF section	Contents
Circulating pressures	DPL, ICP, FCP and K100
Step-down chart	The table and an embedded chart

7. The Two Drawings

- **Well diagram.** A cross-section showing casing, open hole, drill string and surface BOP, with a cyan line for the casing shoe TVD, an orange line for the hole TVD and a green line for the open-hole section.
- **Step-down chart.** Strokes along the bottom and pressure up the side, a green line falling from ICP to FCP, with grid lines every 100 strokes. It appears on screen and in the PDF.

Important note

Kill sheet calculations in a real well-control event must be done by trained, IWCF- or IADC-certified personnel following the operator's procedures and the applicable regulations.